# DEENADAYALAN K

Software Engineer – AI & Data Platforms

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Serving notice – available to join from 29 Dec 2026

## PROFESSIONAL SUMMARY

Software Engineer with 3+ years building an enterprise ETL and data-validation platform in Python, Node.js, FastAPI, Pandas and RabbitMQ. Reconciled 50 million rows with zero differences on a single machine, shipped an OpenAI feature that turns plain-English business rules into Pandas code, and built an MCP server with Claude Code that lets users operate the platform through chat. Promoted to Software Engineer in April 2026; now building agentic, LLM-powered tools on data platforms.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL
**AI / LLM:** OpenAI API, Claude, MCP (Model Context Protocol), LLM Integration, Prompt Engineering, Natural Language to Code, Claude Code
**Backend & APIs:** FastAPI, Node.js, Express.js, FeathersJS, REST APIs, Microservices, WebSockets
**Data Engineering:** ETL Pipelines, Data Reconciliation, Data Validation, Data Quality, Pandas, Dask, RabbitMQ, Node-RED
**Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Apache Hive
**Frontend & Tools:** Angular, React, Git

## PROFESSIONAL EXPERIENCE

**Tristha Global Pvt. Ltd.** – Chennai, India **Sep 2023 – Present**
*Software Engineer (Apr 2026 – Present) · Associate Software Engineer (Sep 2023 – Apr 2026)*
Project: **Travis** – enterprise low-code ETL and data-testing platform

- Built a hash-partitioned reconciliation engine (RabbitMQ queues + Python workers) that scaled from failing under 1 GB to comparing two **75 GB files in ~7 hours** on one 16 GB machine (proof of concept).
- Benchmarked the platform at **50M rows**: loaded CSV into MongoDB at ~52,800 rows/sec and reconciled every row across CSV, DAT, MongoDB and Hive with **zero differences**.
- Shipped an LLM feature (OpenAI API) that converts plain-English business rules into Pandas expressions, so non-developers build transformations without code.
- Built an MCP server for Travis using Claude Code and connected it to the Terra-Sentinel chat interface, so users run platform operations in natural language.
- Fixed RabbitMQ dropping validation workers under load, traced through broker logs; a 50M-row DAT-to-Hive validation went from stalling at 17–74% to **100% complete**.
- Delivered a Python (Pandas, Dask) utility for a client requirement that reconciles DAT and CSV files cell by cell, handling files up to 2 GB.
- Migrated the core transformation engine from Node.js/Formula.js to a Python microservice (Pandas, FastAPI).
- Designed message-driven execution – custom Node-RED nodes wired through RabbitMQ to Python workers – running pipelines independently of the central server and ending bulk-load memory crashes.
- Built a metadata-driven data-quality framework (duplicate, null, format checks) and a General Ledger validation engine that evaluates user-defined rules into Pass/Fail results.
- Compared Dask, Spark, Trino and gRPC against RabbitMQ worker queues for scaling beyond one machine; RabbitMQ was adopted.

**Freelance Software Developer** – Self-employed **Sep 2022 – Aug 2023**

- Built a hospital and patient management system (MongoDB, Express.js, React, Node.js) with REST APIs, authentication and role-based access control.
- Migrated an enterprise Angular application from Angular 8 to 14, rewriting legacy code for newer npm packages and modernising the UI.

## PROJECTS

**DataNerva** – Prompt-Based Data Validation Engine · *Python, LLM* In progress

- Building a validation engine where users describe checks in plain English and the engine runs them, with no SQL or code.

**CuraSynk** – Offline-First Family Health Records App · *Flutter, SQLite, Android* GitHub

- Built and distributed v1.0.1 (signed APK, Play Store closed testing) with family profiles, medical history, medication reminders with local notifications and an offline PDF/image viewer.

## EARLIER EXPERIENCE

**Engineer – Condition Monitoring**, MVS Acmei Technology Pvt. Ltd. Aug 2021 – Aug 2022

- Analysed vibration data from industrial equipment to detect faults and wrote predictive-maintenance reports.

**Graduate Engineering Trainee**, Sri Sai Equipments Pvt. Ltd. Dec 2020 – Jul 2021

**Field Service Technician**, Self-employed Aug 2019 – Jul 2020

## EDUCATION

**B.E. Mechanical Engineering** – Saranathan College of Engineering (Anna University) 2015 – 2019 · CGPA 7.62/10

**Languages:** English, Tamil, Hindi